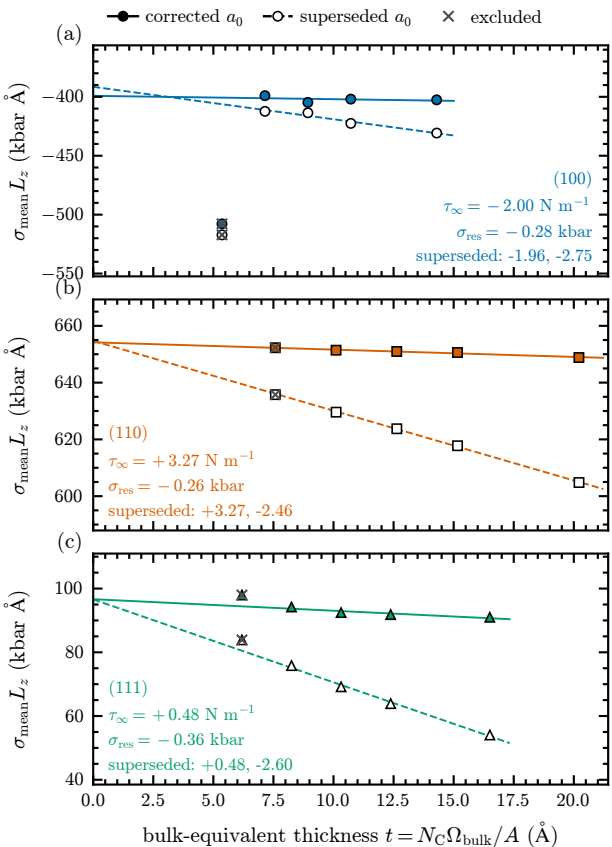




Slab-thickness ladder at both lattice constants. Plotting $\sigma_{\text{mean}} L_z$ against the bulk-equivalent thickness $t = N_C \Omega_{\text{bulk}} / A$ splits the total in-plane force per unit length into a thickness-independent intercept, twice the surface stress τ_{∞} , and a slope σ_{res} carried by the slab interior. t is atom-counted rather than measured as a carbon z-extent because the z-extent depends on where one declares the slab to end: refitting these same points against the z-extent moves τ_{∞} by up to 0.6% here, and up to 4.3% on the superseded ladder, where the larger interior stress amplifies it. Rebuilding the ladder at the corrected $a_0 = 3.572997 \text{ Å}$ (filled, solid) collapses the slope from -2.75...-2.46 kbar at the superseded $a_0 = 3.573641 \text{ Å}$ (open, dashed) to -0.36...-0.26 kbar, while τ_{∞} moves by at most 2.0%. The interior stress was therefore an artefact of the reference lattice constant and not a property of any surface, and removing it left the surface stress intact -- which is what licenses the corrected ladder as the production result. The 6L point (struck through) lies outside the asymptotic regime and is excluded from both fits but still plotted. Each surface keeps its own y-scale because the three sit $\sim 1000 \text{ kbar Å}$ apart. Positive σ means the cell is compressed; τ inherits that sign, so positive τ_{∞} is a compressive surface stress (minus the continuum f). Fits are read from `fit_tau_infinity.py`, not recomputed here.